# Machine Learning 1

Data description

Piotr Wójcik, Szymon Lis, Michał Woźniak

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## Datasets

The data is **exactly the same for all teams**.

### `restaurants` for **classification**

The dataset consists of restaurant records with the following features:

- **restaurant_id** - unique obs identifier
- **has_photo** - Presence of profile photo
- **user_ratings_total** - Cumulative count of user ratings
- **price_level** - Price tier, 1-4 scale (categorical!)
- **category_top20** - Google-defined restaurant category; non-top 20 as “Other”
- **type_meal_takeaway** - Offers takeaway service
- **type_meal_delivery** - Offers meal delivery service
- **type_bar** - Co-located bar operation
- **type_cafe** - Co-located café operation
- **type_night_club** - Co-located night club operation
- **types_count** - Count of co-located operations
- **hours_open** - Total weekly hours open
- **hours_open_weekends** - Total hours open on weekends
- **hours_open_workdays** - Total hours open on workdays
- **days_evenings_only** - Number of days open in the evening hours only
- **days_mornings_only** - Number of days open in the morning hours only
- **weekends_only** - Open only on weekends
- **workdays_only** - Open only on workdays
- **rating_5** - Count of 5-star ratings
- **rating_4** - Count of 4-star ratings
- **rating_3** - Count of 3-star ratings
- **rating_2** - Count of 2-star ratings
- **rating_1** - Count of 1-star ratings
- **rating_avg** - Average rating
- **rating_std** - Standard deviation of ratings
- **tagcat_services** - Number of service-related tags (Wi-Fi, parking, etc.)

- **tagcat_amenities** - Number of amenity-related tags (outdoor seating, etc.)
- **tagcat_atmosphere** - Number of atmosphere-related tags (romantic, casual, etc.)
- **tagcat_offerings** - Number of food/drink offering tags (vegan options, etc.)
- **tagcat_social_inclusivity** - Number of social inclusivity related tags (kid-friendly, etc.)
- **tagcat_payment_options** - Number of payment options (credit card, mobile pay, etc.)
- **place_age_days** - Age of the restaurant in days
- **first_review_year_max** - Year of first review
- **ratings_num_1m_prior** - Number of ratings 1 month prior to survey
- **ratings_num_3m_prior** - Number of ratings 3 months prior to survey
- **ratings_num_6m_prior** - Number of ratings 6 months prior to survey
- **ratings_num_9m_prior** - Number of ratings 9 months prior to survey
- **ratings_num_12m_prior** - Number of ratings 12 months prior to survey
- **ratings_avg_1m_prior** - Average rating 1 month prior to survey
- **ratings_avg_3m_prior** - Average rating 3 months prior to survey
- **ratings_avg_6m_prior** - Average rating 6 months prior to survey
- **ratings_avg_9m_prior** - Average rating 9 months prior to survey
- **ratings_avg_12m_prior** - Average rating 12 months prior to survey
- **lang_pl_count** - Count of reviews in the Polish language
- **rating_pl** - Average rating from reviews in Polish
- **rating_foreign** - Average rating from reviews in languages other than Polish
- **foreign_lang_share** - Proportion of reviews in foreign languages
- **rating_mean_pl** - Mean rating across all Polish reviews
- **rating_mean_foreign** - Mean rating across all foreignlanguage reviews
- **rating_mean_lang_ratio** - Ratio of mean rating in Polish to mean rating in foreign languages
- **review_has_text_pct** - Percentage of reviews that include text
- **review_length_avg** - Average length of review text
- **review_length_std** - Standard deviation of review text lengths
- **catch_restaurant_count_500m** - Number of competing restaurants within 500-meters radius
- **catch_rating_avg_500m** - Average rating of competing restaurants within 500-meter radius
- **catch_place_age_days_500m** - Average age of competing restaurants within 500-meters radius
- **catch_restaurant_count_1000m** - Number of competing restaurants within 1000-meters radius
- **catch_rating_avg_1000m** - Average rating of competing restaurants within 1000-meter radius

- **catch_place_age_days_1000m** - Average age of competing restaurants within 1000-meters radius
- **catch_restaurant_count_2000m** - Number of competing restaurants within 2000-meters radius
- **catch_rating_avg_2000m** - Average rating of competing restaurants within 2000-meter radius
- **catch_place_age_days_2000m** - Average age of competing restaurants within 2000-meters radius
- **residents** - Total population in the 1 km2 census grid cell where the restaurant is located
- **restaurant_count** - Total number of restaurants the 1km2 census grid cell where the restaurant is located
- **restaurants_per_capita** - Number of restaurants per capita in the 1km2 census grid cell where the restaurant is located
- **urbanization** - Degree of urbanization, based on GHS-SMOD layer
- **poi_count_100m** - Number of Points of Interest within 100-meters radius of the restaurant
- **poi_count_200m** - Number of Points of Interest within 200-meters radius of the restaurant
- **poi_count_500m** - Number of Points of Interest within 500-meters radius of the restaurant
- **poi_count_1000m** - Number of Points of Interest within 1000-meters radius of the restaurant
- **poi_count_2000m** - Number of Points of Interest within 2000-meters radius of the restaurant
- **gus_60297** - Number of tourist accommodations in the county
- **gus_60528** - Business deregistrations per capita in the county
- **gus_64428** - Average monthly income in the county
- **gus_79214** - Unemployment rate in the county
- **gus_148074** - Number of foreign tourists in the county
- **gus_153354** - Number of businesses created in the county
- **gus_153398** - Number of businesses deregistered in the county
- **gus_399257** - Utilization of tourist accommodation in the county
- **gus_458173** - Business count per capita in the county
- **gus_1548707_ratio** - New business registrations per capita in the county
- **gus_60528_ratio** - Business deregistrations per capita in the county
- **gus_1548707_net_ratio** - New business registrations net of business deregistrations in the county
- **gus_152173_ratio** - Hospitality business per capita in the county
- **affiliated** - Is a restaurant a part of a chain or franchise with at least 5 restaurants

- **status_closed** - Is a restaurant permanently closed? **DEPENDENT VARIABLE** - only in the training sample

# Files Provided

- `restaurants_train.csv` – training data contains 33296 observations and 86 columns along with the target variable **status_closed**.
- `restaurants_test.csv` – test data contains 8325 observations and 85 columns **without** the target variable.

## `annual.pay.usd` for **regression**

# Target Variable

| Column | Description |
|---|---|
| `annual.pay.usd` | Annual total compensation in USD (salary + bonuses + perks, before taxes). This is the variable you need to predict. |

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## Feature Dictionary

### Demographics & Career

| Column | Type | Description |
|---|---|---|
| `region` | Categorical | Anonymous geographic region code (R01–R18). |
| `age.group` | Ordinal | Age bracket: “18-24”, “25-34”, “35-44”, “45-54”, “55+”. |
| `education` | Ordinal | Highest completed education level (from primary school to professional/doctoral degree). |
| `is.dev.professional` | Categorical | Whether the respondent is a professional developer or codes as part of other work/studies. |
| `employment.type` | Categorical | Primary employment status: Full-time, Part-time, Freelance/Self-employed, Student, Job-seeking. |
| `work.location` | Categorical | Remote, Hybrid, or In-person. |
| `dev.role` | Categorical | Primary developer role (e.g., back-end, front-end, full-stack, mobile, DevOps, data engineer, etc.). |
| `people.manager` | Categorical | Whether the respondent is an individual contributor or a people manager. |

| Column | Type | Description |
|---|---|---|
| `industry` | Categorical | Industry sector of the respondent's employer. |

## Experience

| Column | Type | Description |
|---|---|---|
| `coding.years.total` | Numeric | Total years of coding experience (including education). |
| `coding.years.professional` | Numeric | Years of professional (paid) coding experience. |
| `experience.years` | Numeric | Total years of work experience (any field). |

## Company & Workplace

| Column | Type | Description |
|---|---|---|
| `company.size` | Ordinal | Number of employees at the respondent's company (from freelancer/solo to 10,000+). |
| `tech.purchase.influence` | Ordinal | Level of influence over technology purchasing decisions at work. |
| `build.vs.buy` | Categorical | Preference for building custom solutions vs. buying ready-made products. |
| `cloud.hosting` | Categorical | How the company hosts its applications (cloud, on-premises, hybrid). |
| `first.help.source` | Categorical | Where the respondent goes first for technical help at work. |
| `daily.search.time` | Ordinal | Time spent daily searching for answers to technical problems. |
| `daily.answer.time` | Ordinal | Time spent daily answering technical questions from colleagues. |

## Technical Skills (Multi-Select)

These columns contain **semicolon-separated lists** of technologies. For example: `"JavaScript;Python;TypeScript"`. You will need to parse these to create useful features (e.g., binary indicators for individual technologies, count of technologies used, etc.).

| Column | Description |
|---|---|
| `prog.languages` | Programming languages used professionally in the past year. |
| `databases` | Database systems used in the past year. |
| `cloud.platforms` | Cloud platforms used in the past year. |

| Column | Description |
|---|---|
| web.frameworks | Web frameworks and technologies used in the past year. |
| other.tech | Other frameworks and libraries used in the past year. |
| dev.tools | Developer tools (build, CI/CD, package managers) used in the past year. |
| dev.environments | IDEs and code editors used regularly. |
| personal.os | Operating systems used for personal purposes. |
| work.os | Operating systems used for professional purposes. |
| project.mgmt.tools | Project management and documentation tools used. |
| comm.tools | Communication and collaboration tools used. |

## AI Tool Adoption

| Column | Type | Description |
|---|---|---|
| ai.search.tools | Multi-select | AI-powered search and developer tools used in the past year. |
| ai.tools.used | Multi-select | Specific AI tool use cases (e.g., writing code, searching for answers). |
| uses.ai | Categorical | Whether the respondent currently uses AI tools in development. |
| ai.sentiment | Ordinal | Favorability toward AI tools in the development workflow (from "Very unfavorable" to "Very favorable"). |
| ai.trust | Ordinal | Level of trust in AI tool output accuracy. |
| ai.complex.rating | Ordinal | Rating of how well AI tools handle complex tasks. |
| ai.job.threat | Categorical | Whether the respondent believes AI threatens their job. |

## Learning & Side Projects

| Column | Type | Description |
|---|---|---|
| how.learned.coding | Multi-select | How the respondent learned to code (bootcamp, university, self-taught, etc.). |
| side.coding | Multi-select | Types of coding activities done outside of work (hobby, open source, freelance, etc.). |

## Job Satisfaction

| Column | Type | Description |
|---|---|---|
| job.satisfaction | Numeric (0–10) | Overall satisfaction with current professional role. |

## Important Notes

1. **Missing values**: Many columns contain missing values (NaN). Handling these appropriately is part of the task. Consider whether missingness itself might be informative.

2. **Multi-select columns**: Columns like `prog.languages`, `databases`, etc. contain semicolon-separated lists. You need to decide how to encode these – common approaches include:

    - Binary indicator columns for the most frequent items
    - Count of items selected
    - TF-IDF or frequency-based encoding

3. **Target variable distribution**: The salary distribution is right-skewed. Consider whether a log transformation of the target improves your model.

4. **Outliers**: Some salary values may be unusually low or high. Investigate and decide how to handle them.

5. **Ordinal variables**: Several features have a natural ordering (age, education, company size, experience). Think about whether to treat them as numeric or categorical.

6. **Region effects**: The `region` column captures geographic differences in compensation. Different regions have substantially different salary levels.

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## Files Provided

- `train.csv` – training data contains 2,512 observations and 41 columns along with the target variable **annual.pay.usd**.
- `test.csv` – test data contains 628 observations and 41 columns (40 features + `id`) **without** the target variable.
- `sample_submission.csv` – sample submission file in the correct format.